

DISCOVERY, DISASTERS, AND DREAMS

Bridging LA's Water History

ST

Dennis Kopp

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DISCOVERY, DISASTERS, AND DREAMS Dennis Kopp

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DISCOVERY, DISASTERS, AND DREAMS

Bridging LA's Water History

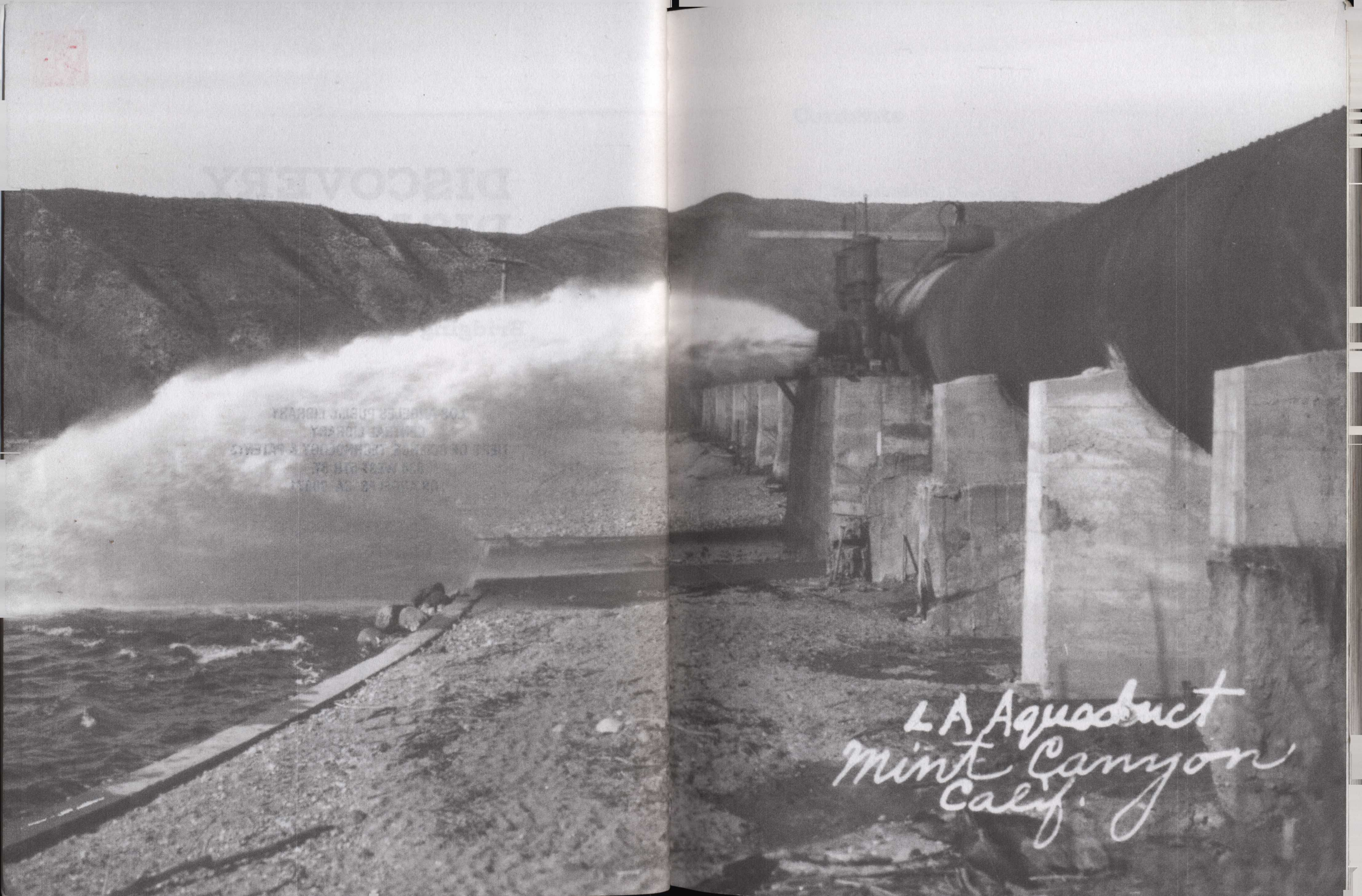
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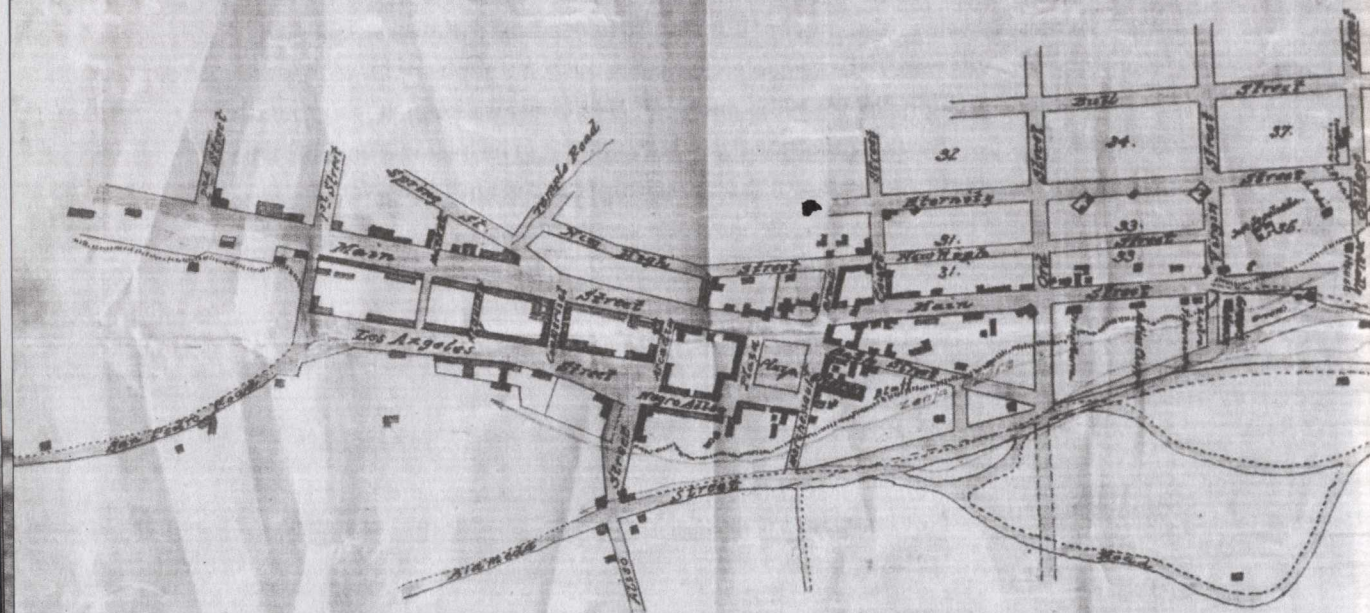
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LA Aqueduct
Mint Canyon
Calif.

Wm. Moore, 1868.



Map showing the route of the Zanja Madre irrigation system, 1888.
(Security Pacific National Bank Collection)

“He who brings the water will bring the people.”

—William Mulholland

Aqueducts, or guided waters, have existed throughout the history of the ancient world. But it is the Romans and their system of aqueducts, built to supply Rome's population with fresh water, the world most closely associates with these life-sustaining structures. Arguably, were it not for its aqueducts and its effective distribution of water, Rome's historical achievements would not have been possible.

In the late first century AD, a prominent Roman civil engineer named Sextus Julius Frontinus was the water commissioner under emperor Nerva. He wrote an official report, *De aquis urbis Romae*, for the emperor on the status of Rome's aqueducts, meticulously mapping the water-supply system, assessing its condition, and looking for loss or leakage through either neglect or thievery. Water distribution depended to a certain extent on its height as it entered the walled city. The other factor was purity. The more potable was segregated into a system for drinking. The least desirable was sent through the *Cloaca Maxima* or sewer which eventually led to the Tiber; an undignified tribute to the river which gave safe passage to *Romulus and Remus*.

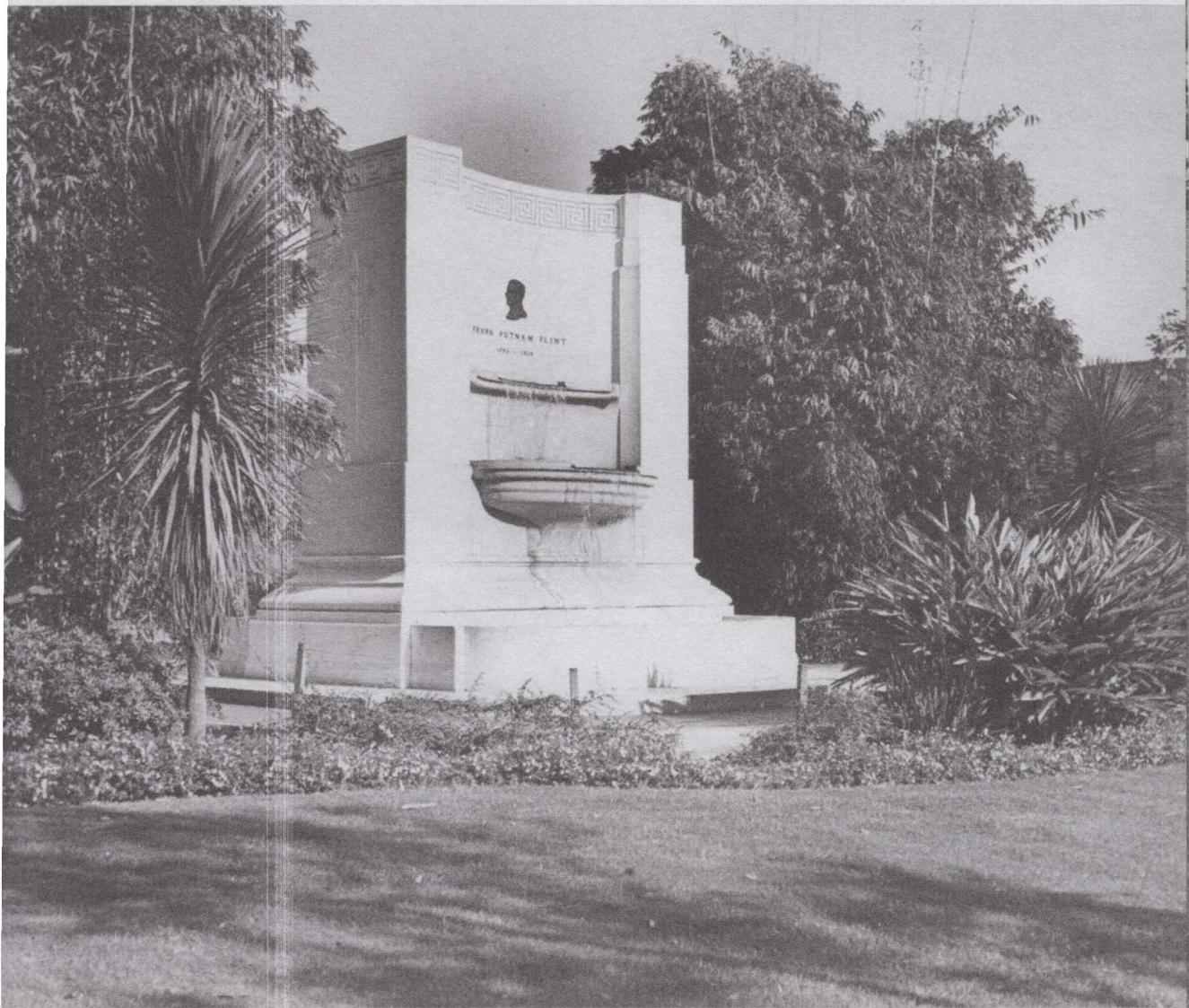
As Rome had its Frontinus, so too did Los Angeles. William Mulholland was born in 1855 in Ireland. As a teenager, he became a British merchant seaman sailing back and forth across the Atlantic. After concluding that a life at sea wouldn't add up to much more than a life at home, he made his way to New York City in 1874. With his younger brother, Hugh Jr., Mulholland arrived in the dusty pueblo of Los Angeles by horseback from San Francisco. (The pair had been caught as stowaways on a ship sailing from New York to California with their uncle's family. They were dropped off at the Isthmus of Panama. Without money for train fare, the brothers hiked the 47

Opposite: William Mulholland standing in the outdoors. (Security Pacific National Bank Collection)

Overleaf: Construction of ditches in the foreground, and the workmen's camp in the background, 1912. (Security Pacific National Bank Collection)

Overleaf Inset: Portrait of William Mulholland, who brought water to Los Angeles from the Owens Valley. Featured in the book, *La Reina*, by Laurance Landreth Hill, published in 1929. (Security Pacific National Bank Collection)





View of the Frank Putnam Flint Memorial Fountain on the south lawn of Los Angeles City Hall, Jan. 1937. The Flint Fountain, which uses marble from the same quarry used for the Lincoln Memorial, is a civic landmark honoring former U.S. Senator Frank Putnam Flint. Sadly, Senator Flint died just before the fountain was dedicated. Senator Flint, who served from 1905-1911, was a strong advocate in favor of building the Los Angeles Aqueduct system and bringing water from the Owens Valley to Los Angeles. (Works Progress Administration Collection)

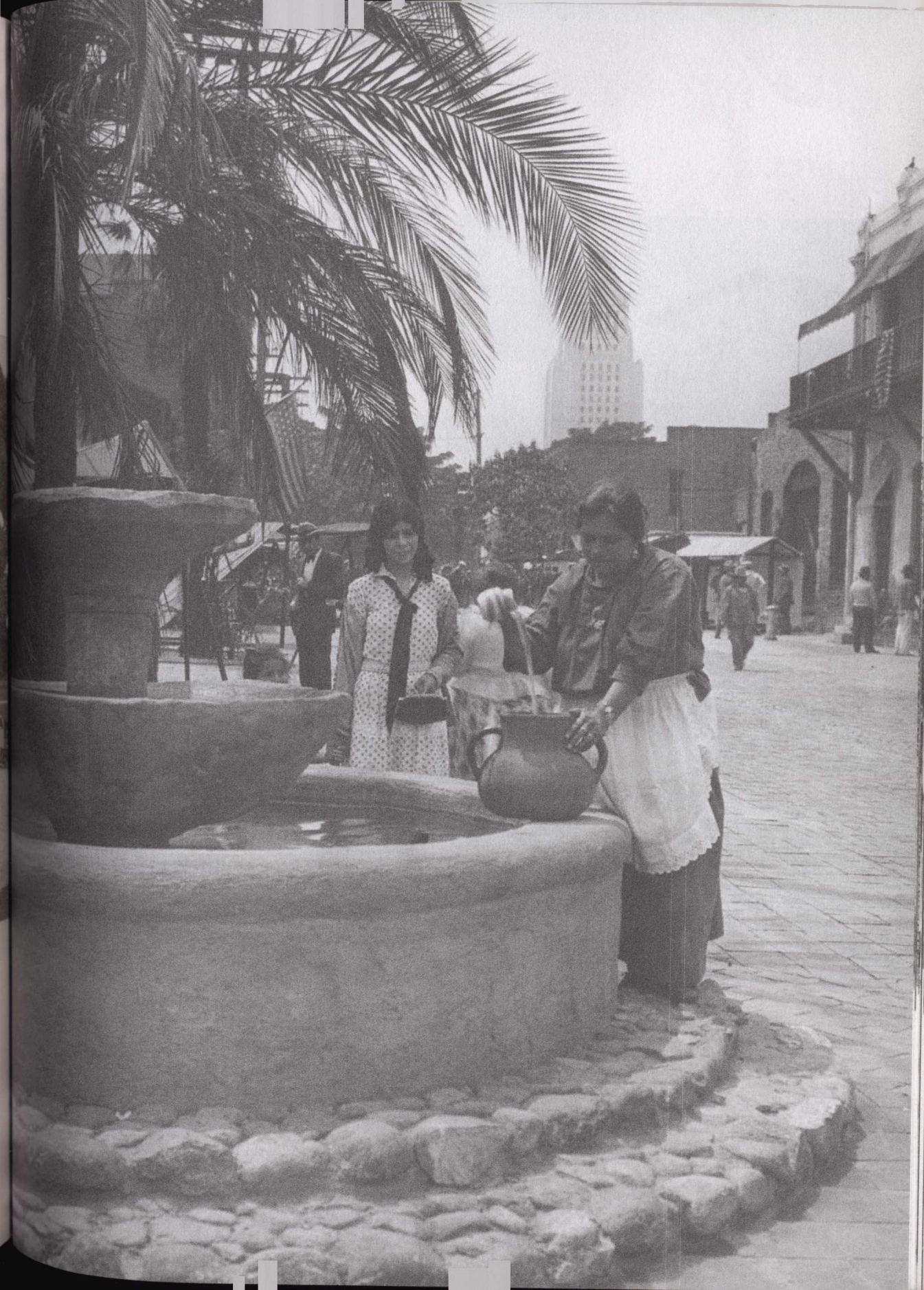


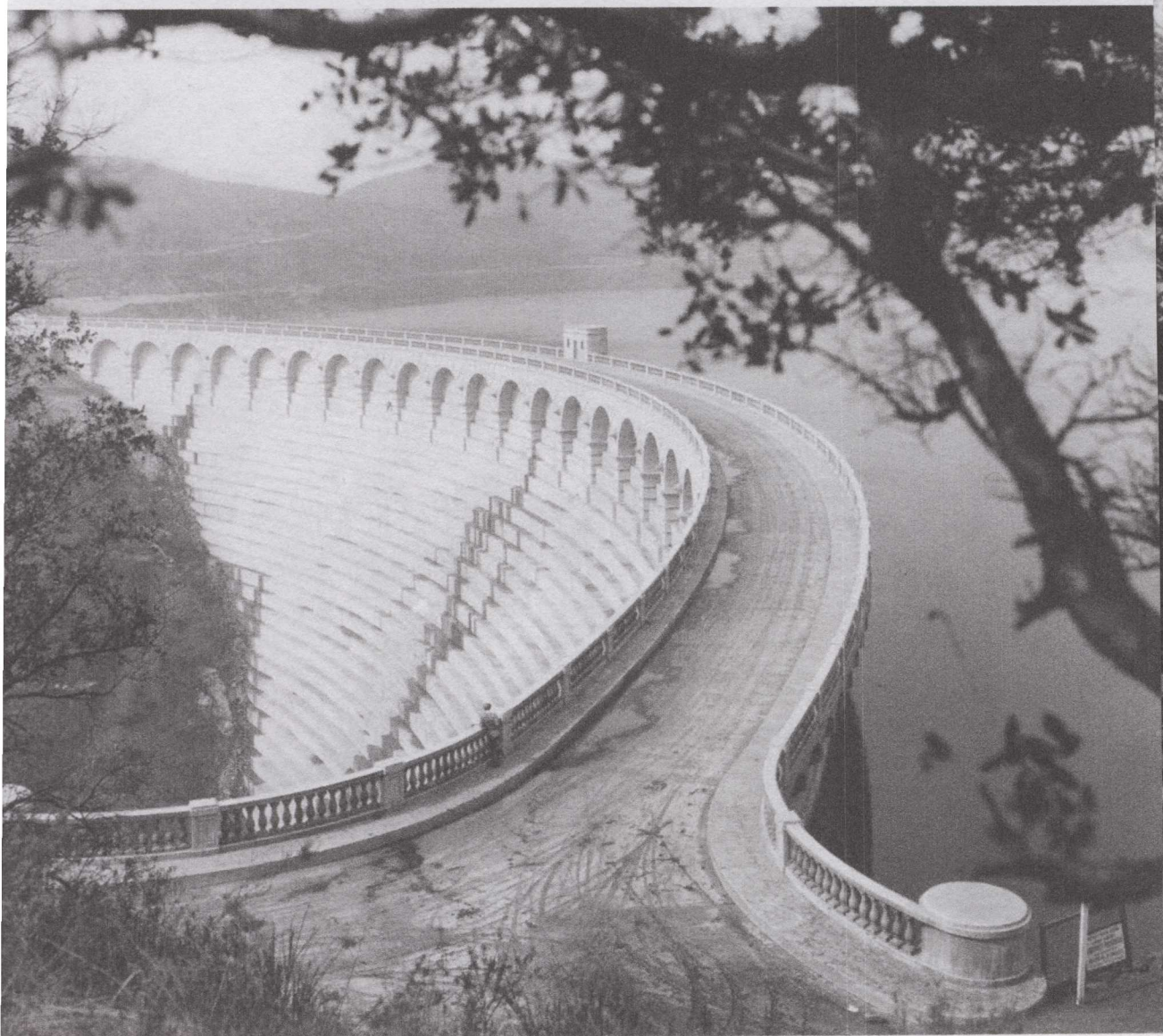
Fountain in Pershing Square, looking northwest, 1928. (Security Pacific National Bank Collection)



Several people are gathered around or in the circular water fountain in South Park. Beyond it lies a road lined with high palm trees and an equally lone line of street lamps, circa 1962. (Security Pacific National Bank Collection)

Opposite: One woman fills a pot with water from the fountain, while another woman with a child watches her. City Hall is partially visible in the background. (Security Pacific National Bank Collection)





View of Mulholland Dam in the Hollywood Hills, the most beautiful of a score of storage basins in Los Angeles' water system, circa 1930. (Herald Examiner Collection)



Hollywood Reservoir and Mulholland Dam during dedication ceremonies, showing dignitaries in a car with Los Angeles City seal on the door. To the left are flags and the edge of a dedicatory plaque can be seen. (Security Pacific National Bank Collection)

“The Los Angeles River is now a mammoth concrete drainage ditch. Taming a river is one thing... Emasculating it is another.”

—Dick Roraback, *Los Angeles Times* (Oct. 1985)

Starting in 1917, Congress had been appropriating funds for flood control measures initially along the Mississippi, Ohio, and Sacramento rivers. Under authorization of the 1936 and 1938 Flood Control Acts, construction of the Los Angeles River channel began in 1936. Work to completely channelize the river began a few months following the 1938 flood.

By the end of the project in 1959, the U.S. Army Corps of Engineers moved more than “20 million cubic yards of earth (equivalent to roughly 800,000 dump trucks), mixed 3.5 million barrels of concrete, placed 147 million pounds of reinforced steel, and set 460,000 tons of stones,” according to Blake Grumpecht’s *The Los Angeles River, Its Life, Death and Possible Rebirth*.

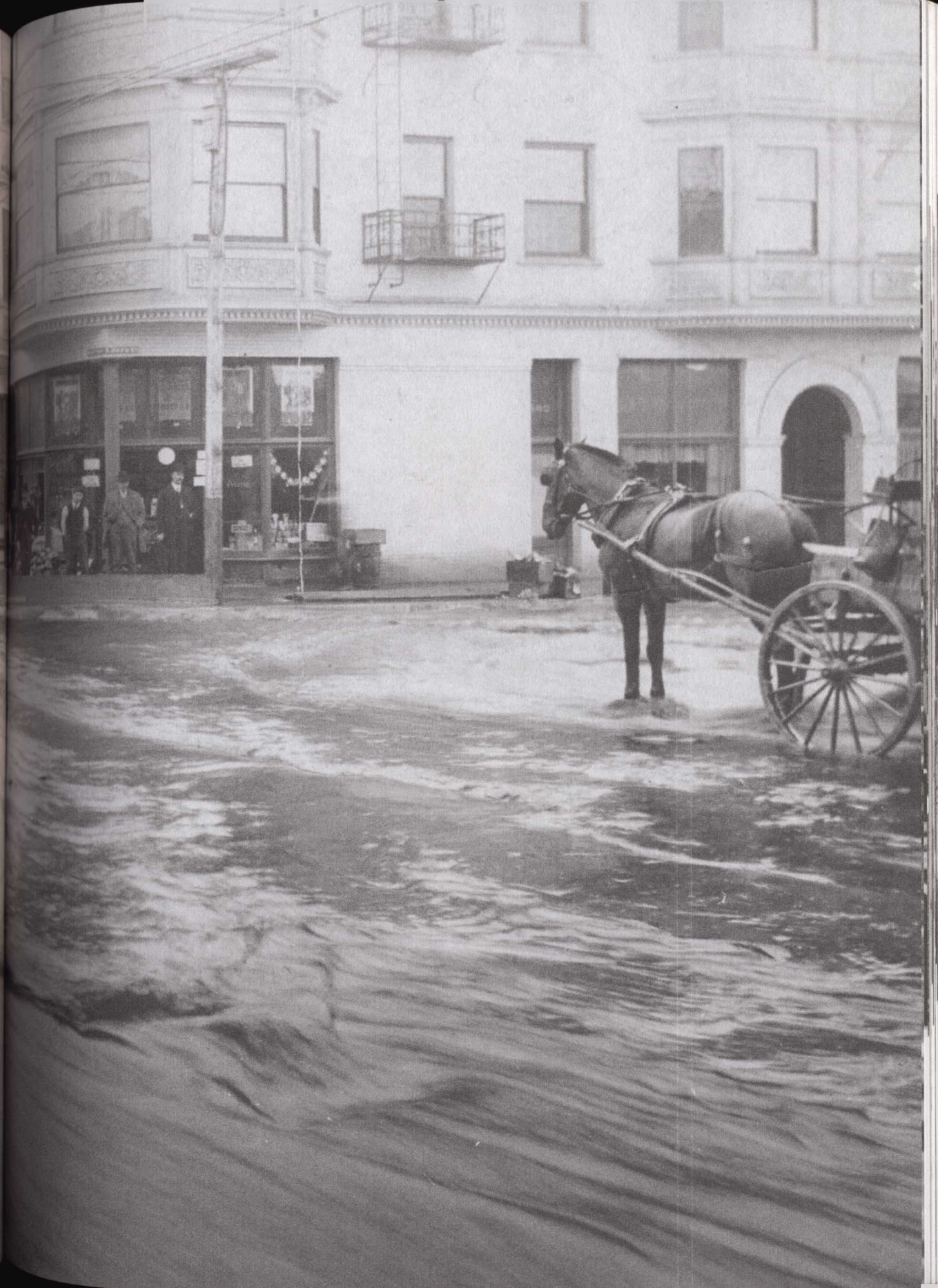
More than 300 bridges were built and 278 miles of river and tributaries were retrofitted. By connecting streets and sewers to drains along the river, rainfall could be quickly captured from the surrounding streets.¹ Upon completion of the Los Angeles River Channel, it was said “A single drop of rain falling high in the San Gabriel Mountains can now reach the sea in less than sixty minutes.”

Opposite: “From Universal City to Studio City, Army engineers are creating a new course for the oceanbound water from the Valley and the hills around the Valley’s rim. When the engineer’s work is done, the Los Angeles River will be enclosed in steel and concrete all the way to the Sepulveda Dam in Van Nuys and the western branch of the Tujunga Wash, shown at its confluence with the river. It will be channeled as far up as Hansen Dam near Roscoe,” 1948. The Republic Studios can be seen at the bottom left of the photograph and Colfax Ave. at top right. (Valley Times Collection)

Overleaf: “Flood control experts have condemned the once-rampaging Los Angeles River to life imprisonment in a concrete-walled channel like that shown in Universal City. Before the channel was built, the river meandered quietly through a shrub-choked course to burst into life at sudden intervals when ‘flash-floods’ swept through it, cutting away the banks, inundating nearby areas, and smashing highway bridges like kindling wood,” 1949. (Valley Times Collection)



The flooded intersection of Hope and 6th Streets,
looking southeast, during flooding in 1905.
(Security Pacific National Bank Collection)

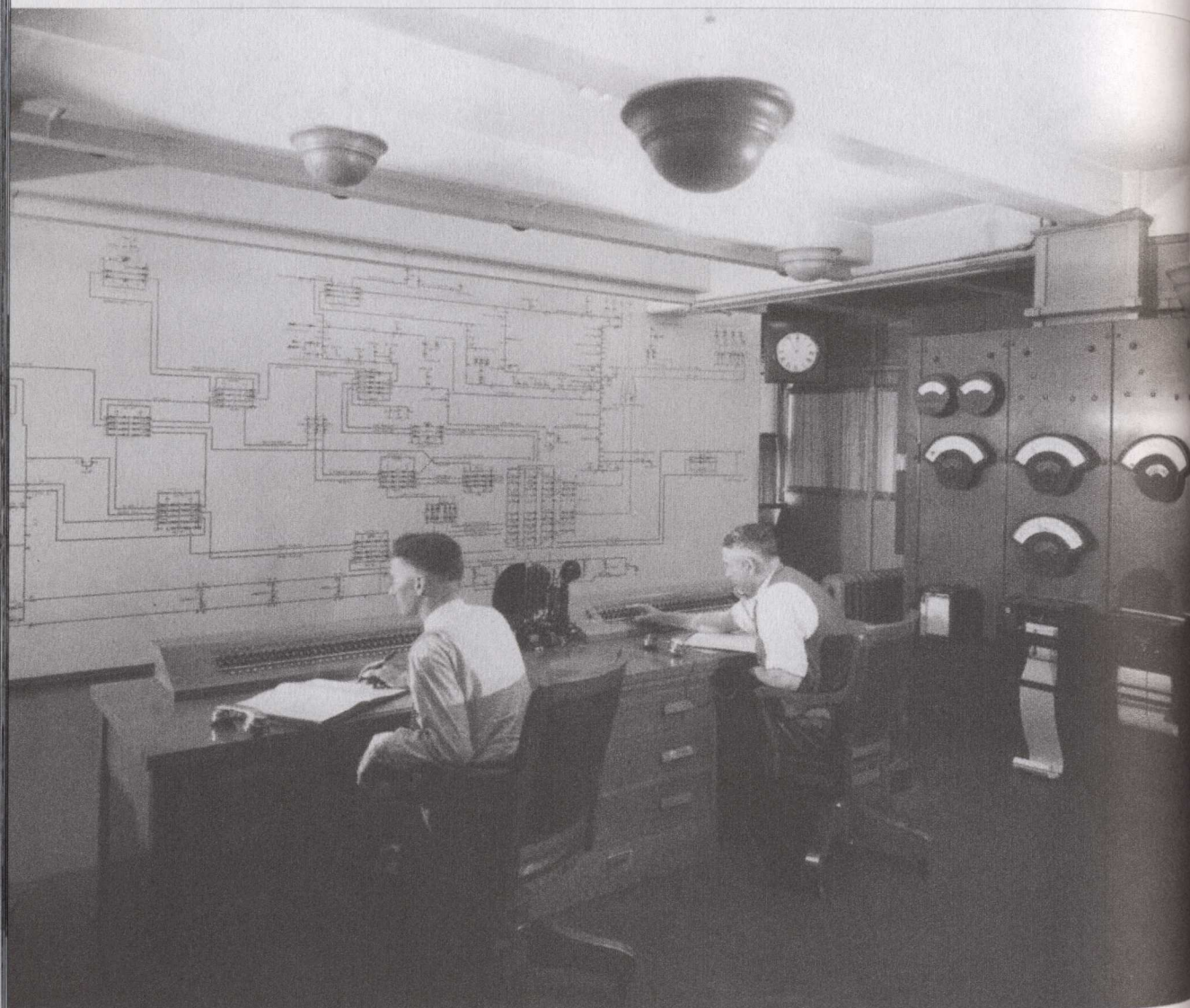




Power house #2 and the crew's tree-surrounded clubhouse below the St. Francis Dam as they were before the break. (Los Angeles Herald Examiner Collection)



Two generators, the crumbled foundations of the power house, a pool of water, and a mudhole remained, a sink of death where the clubhouse stood before. (Los Angeles Herald Examiner Collection)

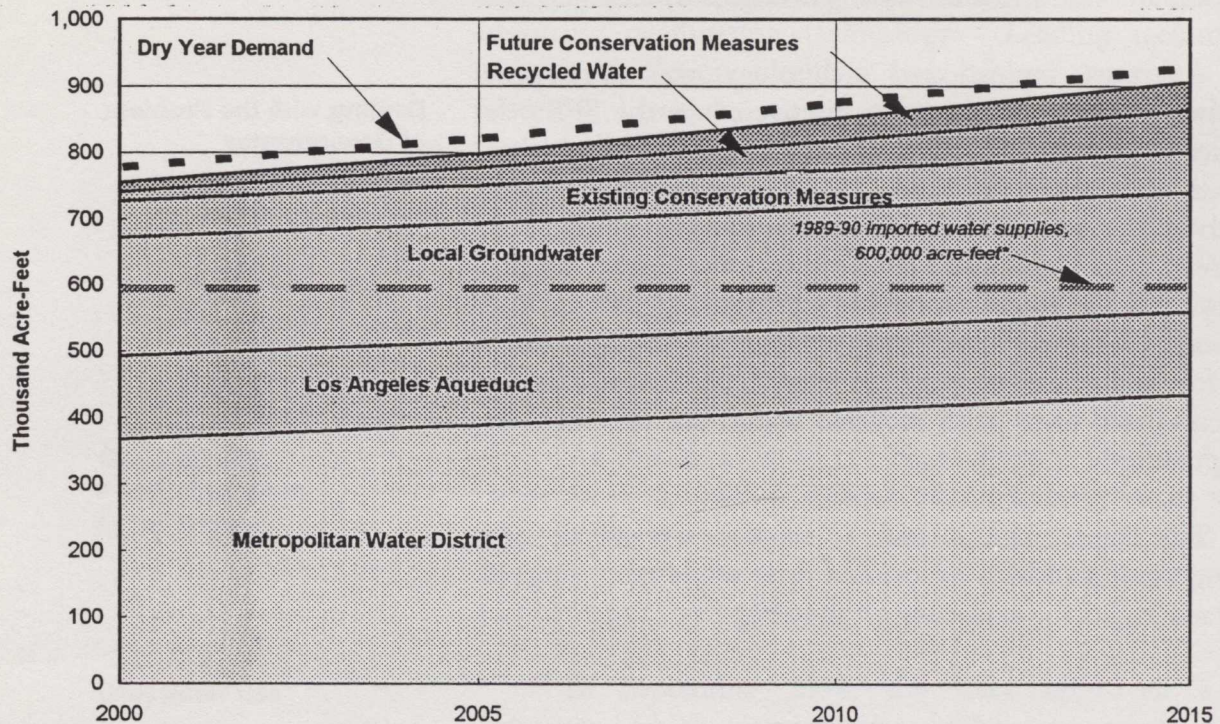


Two men work at a long desk in the control room for the LA Aqueduct. In front of them is a diagram filling the entire wall. One man reaches for the control buttons in front of him on the desk. Both have telephone hand sets laying on their desk, 1938. (Security Pacific National Bank Collection)



The LA Aqueduct flows south between the mountains and Antelope Valley in this aerial view, 1983. (Herald Examiner Collection)

BALANCING SUPPLY AND DEMAND TO FULFILL LOS ANGELES' FUTURE WATER NEEDS IN A DRY YEAR



Projected Future Dry Year Scenario Between Supply and Demand	2000	2005	2010	2015
	(all values in acre-feet)			
Los Angeles Aqueducts (1)	125,000	125,000	125,000	125,000
Local Wells (2)	180,000	180,000	180,000	180,000
MWD (3)	367,000	388,000	411,000	435,000
Conservation Measures (up to 1995)	55,000	57,000	58,000	61,000
Total Traditional Supplies	727,000	750,000	774,000	801,000
<i>Additional Conservation Measures</i>	<i>13,000</i>	<i>27,000</i>	<i>44,000</i>	<i>62,000</i>
<i>Recycled Water Yield (4)</i>	<i>15,000</i>	<i>21,000</i>	<i>34,000</i>	<i>43,000</i>
Total Available Supply	755,000	798,000	852,000	906,000
Total Demand (5)	787,000	827,000	878,000	928,000
<i>Deficit (6)</i>	<i>32,000</i>	<i>29,000</i>	<i>26,000</i>	<i>22,000</i>

Notes: * 1989-90 imported supplies from the LAA and MWD is for the third year of a six year drought.

It is shown to provide a basis of comparison for expected imported supplies in the future.

(1) LAA supplies are based on lowest deliveries of record (July 1, 1990 - June 30, 1991).

(2) Local well production includes water recycled and used for groundwater recharge.

(3) MWD supplies are based on their 1995 Drought Contingency Plan, and is assumed to keep pace with a projected annual population growth rate of 1.14 percent.

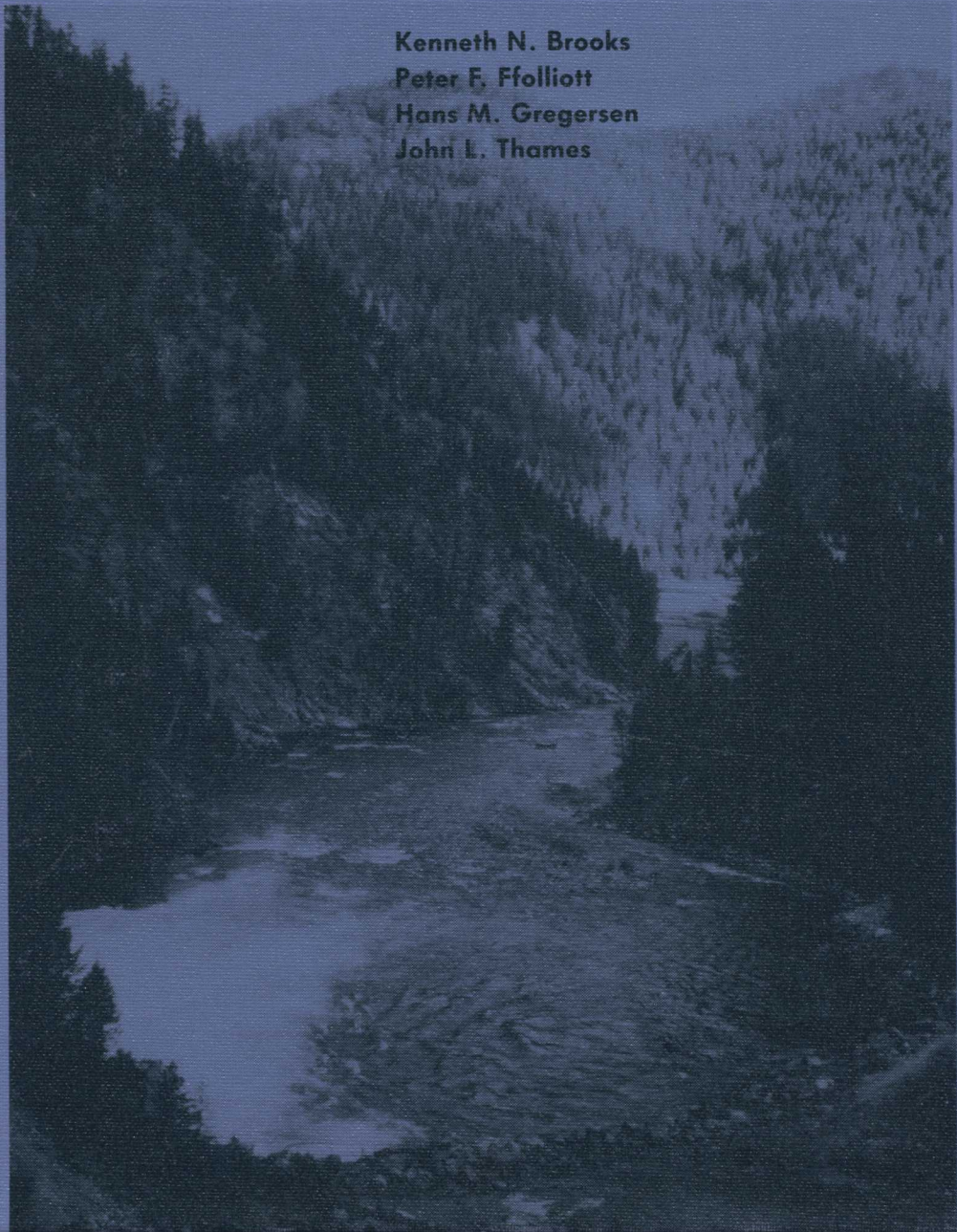
(4) Recycled water supplies are based on irrigation and industrial use only. It excludes the portion produced for groundwater recharge which is included in the local well production.

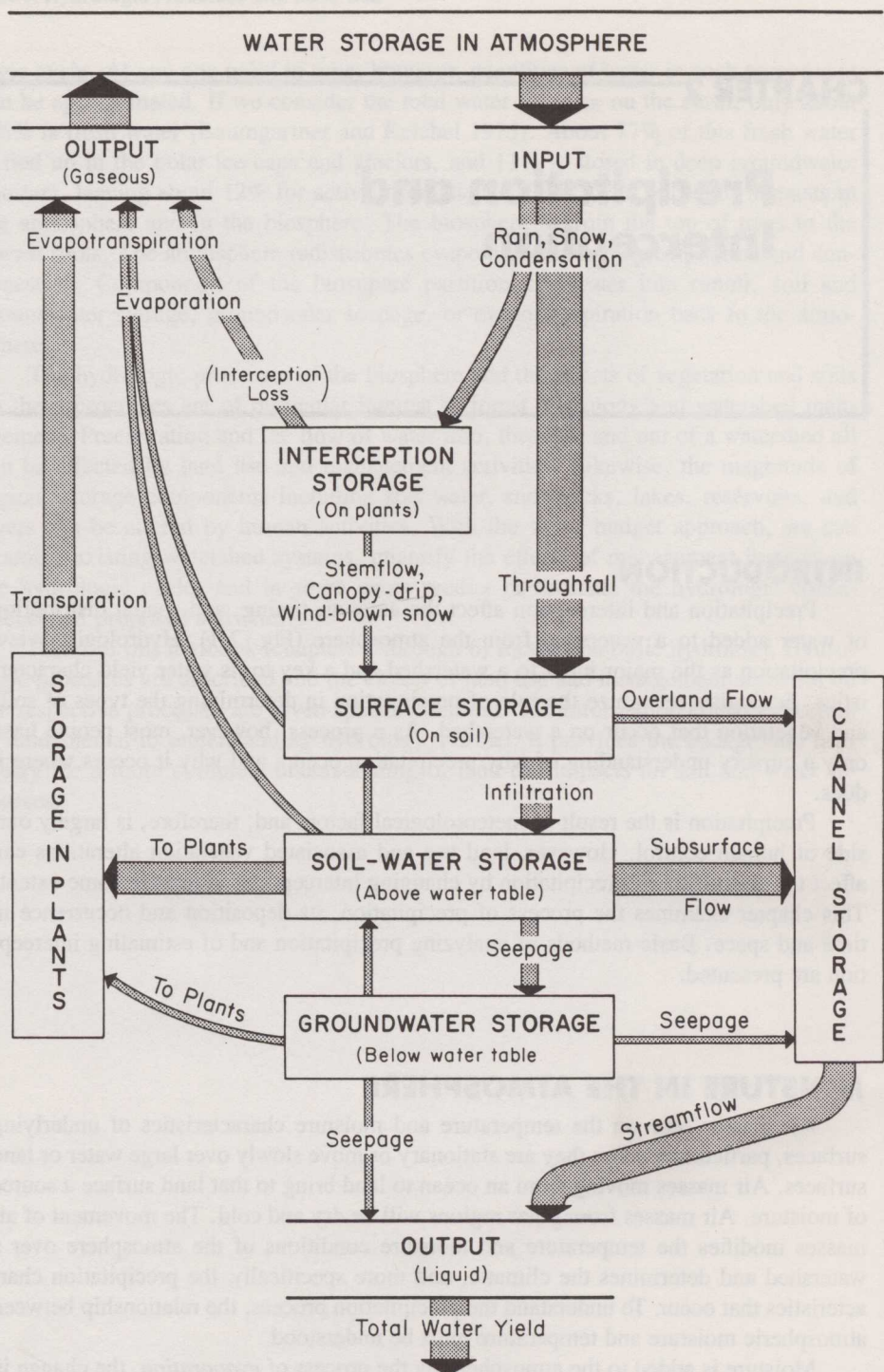
(5) Dry year demand values are obtained by increasing normal demand values by 6 percent.

(6) Projected supply deficits will be made up with increased conservation efforts by City customers.

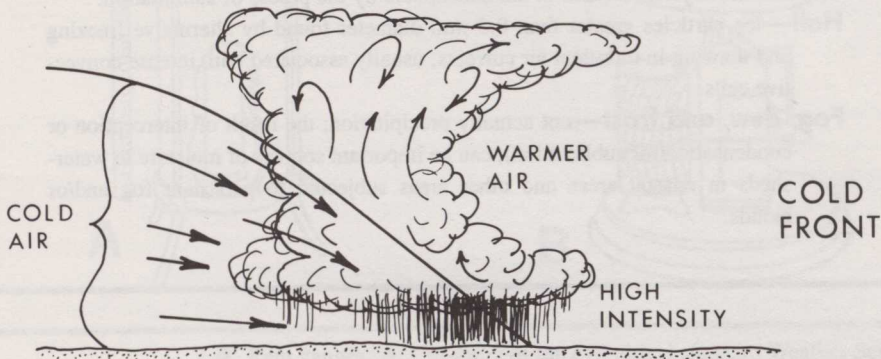
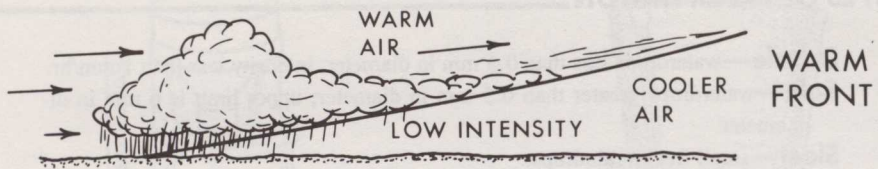
HYDROLOGY AND THE MANAGEMENT OF WATERSHEDS

Kenneth N. Brooks
Peter F. Ffolliott
Hans M. Gregersen
John L. Thames





2.1. The hydrologic cycle consists of a system of water-storage compartments and the solid, liquid, or gaseous flows of water within and between the storage points (from Anderson et al. 1976).



FRONTAL SYSTEMS

WINDWARD SIDE

LEEWARD SIDE

LIFTING
AND
COOLING

WARMING
AND
DRYING

RAIN
SHADOW

OROGRAPHIC PRECIPITATION

SOLAR
RADIATION

ICE CRYSTALS

TURBULENCE
AND
INSTABILITY

HIGH TEMPERATURES
AND MOISTURE

CONVECTIVE PRECIPITATION

2.3. Examples of different mechanisms causing air masses to lift and cool, resulting in precipitation.

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ARROYO SECO WATERSHED RESTORATION FEASIBILITY STUDY

VOLUME I PROJECT REPORT

May 31, 2002

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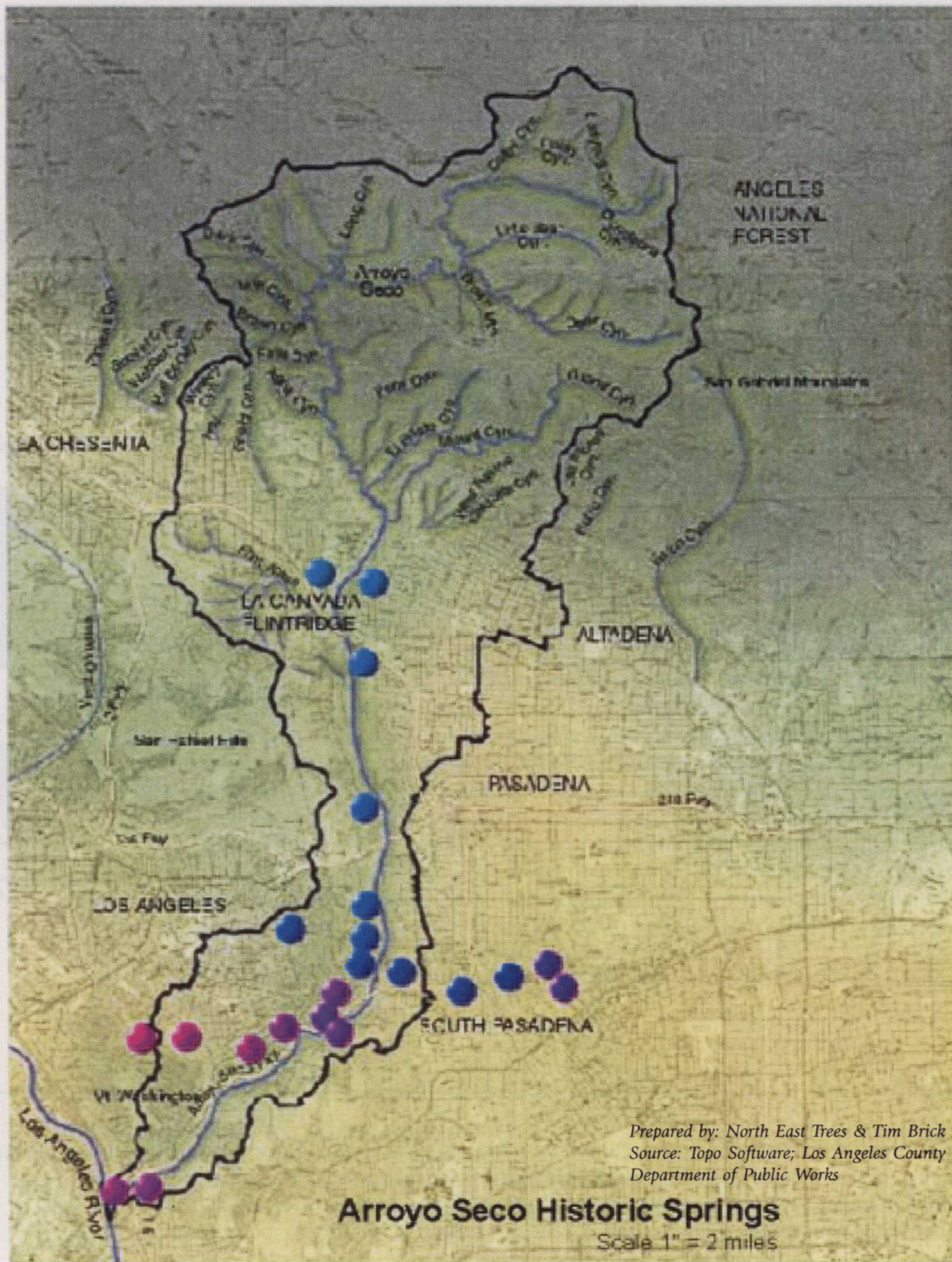


FIGURE II-3: ARROYO SECO HISTORIC SPRINGS

Arroyo Seco Watershed Restoration Feasibility Study

TABLE IV- I: FRAMEWORK FOR RESTORATION PROJECTS: WATERSHED RESTORATION GOALS, GUIDELINES & ACTION ITEMS (CONT.)

RESTORATION GOAL 4: IMPROVE RECREATIONAL OPPORTUNITIES AND ENHANCE OPEN SPACE			
4.1	Improve connectivity and public access from the Angeles National Forest to the coastal shore.	4.1.1	Improve the network of trails by connecting trails and open spaces with multi-modal trails.
		4.1.2	Provide access for public use of the watershed's rivers and streams.
4.2	Protect and interpret natural, community, cultural and historic resources.	4.2.1	Improve connectivity between parks, open space, cultural and historic resources through trails and land acquisition.
		4.2.2	Develop trails that connect residents across communities to meaningful destinations.
4.3	Integrate natural resources management with recreational needs.	4.3.1	Manage parks and other open spaces, including golf courses, trail connections and native habitat to benefit flood management.
		4.3.2	Provide opportunities for a range of recreational activities.
		4.3.3	Coordinate between agencies and groups to meet community needs for recreation with agency priorities
4.4	Protect existing open space while augmenting open space network	4.4.1	Inventory existing and potential open spaces for protection, recreation, land use conversion.
		4.4.2	Introduce the concept of long-term acquisition with an open public process.
4.5	Improve visual quality of the landscape.	4.5.1	Expand urban tree-planting programs.
4.6	Mediate conflicts between recreation and conservation and opposing recreational users.	4.6.1	Improve the cooperative spirit among recreational users through an open public process.

D. RECOMMENDATIONS FOR PROJECTS AND PROGRAMS

Through the course of the planning process, the Project Team identified over 78 potential projects, programs, and best management practices. These ranged from a redesign of a golf course to the daylighting of streams watershed-wide. The matrix represents some possible projects, but this list is by no means comprehensive. A matrix was developed to rank the potential benefits of each site for the different specializations represented in the study team: hydrology, water resources & quality, habitat, and recreation. Figure IV-1: *Arroyo Seco Watershed Restoration Plan-Locations of Projects and Programs*, identifies the location of site-specific projects and programs.

This matrix is intended for city, county, regional, state and federal agencies (hereinafter referred to as "agencies") with jurisdictions within the watershed, and for organizations that focus on issues in this region. The first step to using this matrix is for an agency to identify which projects are appropriate to undertake per their jurisdiction. Next, appropriate projects need to be prioritized according its ability to meet an agency's goals and work within a realistic budget and schedule. Often a project will meet an agency's needs, but funding will not be available to implement that desired project. If this is the case, that agency can seek out funding sources, such as Proposition 12, 13 or 40 funds from the state. Assistance on these grant proposals is available from the project team or other entities.

The structure of the matrix is simple. Projects and programs (hereinafter referred to as "projects") occur on each row. They are grouped according to sub regions within the watershed, such as "I. Angeles National Forest" or "VII. Arroyo through Los Angeles." There are a total of eight subregions, generally falling within the Angeles National Forest, foothill communities of La Cañada Flintridge and Altadena, cities of Pasadena, South Pasadena and Los Angeles.

Each project is given a title for ease of communication. These are suggestions only. An organization may see the usefulness in combining several projects into a single effort if